GO CALCULATOR — CANONICAL EXECUTION SPECIFICATION

Version: 1.0.0

Status: Authoritative

Purpose: Complete deterministic specification of the Outcome Resolution Engine

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1. SYSTEM BOUNDARY

Layer 1 (Intent Parsing): LLM-based, non-authoritative, converts natural language to structured intent.

Layer 2 (Outcome Resolution Engine): THIS SPECIFICATION. Deterministic, mathematical, no AI, no inference.

Guarantee: Given identical inputs, outputs are byte-identical across all executions.

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2. INPUT SCHEMAS

2.1 Intent Object (from Layer 1, untrusted)

```typescript
```

```typescript
interface Intent {
  targetEffects: {
    energy: number;        // [-1.0, 1.0] where -1=sedation, 0=neutral, 1=stimulation
    focus: number;         // [0.0, 1.0] where 0=unfocused, 1=laser focus
    mood: number;          // [-1.0, 1.0] where -1=introspective, 0=neutral, 1=euphoric
    body: number;          // [0.0, 1.0] where 0=no body effect, 1=heavy body load
    creativity: number;    // [0.0, 1.0]
  };
  constraints: {
    maxAnxiety: number;    // [0.0, 1.0] HARD LIMIT, default 0.3
    minTHC?: number;       // percentage, optional
    maxTHC?: number;       // percentage, optional
    minCBD?: number;       // percentage, optional
    maxCBD?: number;       // percentage, optional
  };
  context: {
    timeOfDay?: "morning" | "afternoon" | "evening" | "night";
    tolerance?: "low" | "medium" | "high";
    experience?: "beginner" | "intermediate" | "expert";
  };
}
```

Validation Rules:

- All `targetEffects` values MUST be within specified ranges
- If outside bounds, clamp to nearest valid value
- `maxAnxiety` defaults to 0.3 if not specified
- Cannabinoid constraints are optional; if specified, must be [0, 100]

2.2 Inventory Schema

```
typescript
```

```
interface Cultivar {
  id: string;                        // unique identifier
  name: string;                       // display name
  thcPercent: number;                 // [0, 100], from COA
  cbdPercent: number;                 // [0, 100], from COA
  terpenes: Record<string, number>;   // terpene_name -> percentage [0, 100]
  available: boolean;                 // in stock
  grower?: string;                    // optional metadata
  batch?: string;                     // optional metadata
}

interface Inventory {
  cultivars: Cultivar[];
  timestamp: string;                  // ISO 8601, for audit only
}
```

Validation Rules:

- `thcPercent + cbdPercent` may exceed 100 (total cannabinoids can vary)
- Terpene keys are case-insensitive, normalized to lowercase
- Terpene percentages summing >10% triggers warning but does not invalidate
- Only `available: true` cultivars are considered

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3. EFFECT DIMENSIONS

3.1 Canonical Dimensions

Dimension	Range	Interpretation
`energy`	[-1.0, 1.0]	-1.0 = maximum sedation, 0.0 = neutral, +1.0 = maximum stimulation
`focus`	[0.0, 1.0]	0.0 = scattered/unfocused, 1.0 = laser focus
`mood`	[-1.0, 1.0]	-1.0 = introspective/contemplative, 0.0 = neutral, +1.0 = euphoric
`body`	[0.0, 1.0]	0.0 = cerebral only, 1.0 = heavy body sensation
`creativity`	[0.0, 1.0]	0.0 = analytical, 1.0 = divergent thinking
`anxiety`	[0.0, 1.0]	0.0 = no anxiety risk, 1.0 = high anxiety risk (CONSTRAINT ONLY)

4. TERPENE INFLUENCE FRAMEWORK

4.1 Terpene-to-Effect Mapping (MODELED COEFFICIENTS)

Critical Note: These are MODELED INFLUENCE COEFFICIENTS, not measured effects. They represent the designed directional influence of each terpene on effect dimensions.

```
TERPENE_INFLUENCES = {
  // Primary monoterpenes
  "limonene": {
    energy: 0.6, focus: 0.3, mood: 0.8, body: 0.0, creativity: 0.5, anxiety: -0.4
  },
  "pinene": {
    energy: 0.4, focus: 0.7, mood: 0.2, body: 0.0, creativity: 0.3, anxiety: -0.3
  },
  "myrcene": {
    energy: -0.7, focus: -0.3, mood: 0.1, body: 0.8, creativity: 0.0, anxiety: -0.5
  },
  "linalool": {
    energy: -0.4, focus: 0.0, mood: 0.3, body: 0.3, creativity: 0.2, anxiety: -0.7
  },
  "caryophyllene": {
    energy: 0.0, focus: 0.1, mood: 0.2, body: 0.5, creativity: 0.0, anxiety: -0.6
  },
  "humulene": {
    energy: -0.2, focus: 0.2, mood: 0.0, body: 0.3, creativity: 0.1, anxiety: 0.0
  },
  "terpinolene": {
    energy: 0.5, focus: 0.2, mood: 0.6, body: 0.1, creativity: 0.7, anxiety: 0.1
  },
  "ocimene": {
    energy: 0.3, focus: 0.1, mood: 0.5, body: 0.0, creativity: 0.4, anxiety: -0.2
  },

  // Secondary terpenes
  "bisabolol": {
    energy: -0.3, focus: 0.0, mood: 0.2, body: 0.2, creativity: 0.0, anxiety: -0.5
  },
  "camphene": {
    energy: 0.2, focus: 0.3, mood: 0.0, body: 0.1, creativity: 0.0, anxiety: 0.0
  },
  "valencene": {
    energy: 0.3, focus: 0.2, mood: 0.4, body: 0.0, creativity: 0.3, anxiety: -0.1
  },
  "geraniol": {
    energy: 0.1, focus: 0.0, mood: 0.4, body: 0.1, creativity: 0.3, anxiety: -0.3
  },
  "nerolidol": {
    energy: -0.5, focus: -0.1, mood: 0.2, body: 0.4, creativity: 0.1, anxiety: -0.4
  },
```

```
  "borneol": {
    energy: -0.3, focus: 0.1, mood: 0.0, body: 0.3, creativity: 0.0, anxiety: -0.2
  },
  "eucalyptol": {
    energy: 0.3, focus: 0.5, mood: 0.1, body: 0.0, creativity: 0.2, anxiety: 0.0
  },
  "pulegone": {
    energy: 0.1, focus: 0.4, mood: 0.0, body: 0.0, creativity: 0.1, anxiety: 0.1
  },
  "sabinene": {
    energy: 0.2, focus: 0.2, mood: 0.1, body: 0.1, creativity: 0.1, anxiety: 0.0
  },
  "terpineol": {
    energy: -0.4, focus: 0.0, mood: 0.2, body: 0.3, creativity: 0.1, anxiety: -0.4
  },
  "carene": {
    energy: 0.1, focus: 0.3, mood: 0.0, body: 0.2, creativity: 0.0, anxiety: 0.0
  },
  "cymene": {
    energy: 0.2, focus: 0.1, mood: 0.1, body: 0.0, creativity: 0.1, anxiety: 0.1
  },
  "fenchol": {
    energy: -0.2, focus: 0.0, mood: 0.1, body: 0.2, creativity: 0.0, anxiety: -0.3
  },
  "guaiol": {
    energy: -0.3, focus: 0.0, mood: 0.1, body: 0.3, creativity: 0.0, anxiety: -0.2
  },
  "isopulegol": {
    energy: 0.0, focus: 0.2, mood: 0.0, body: 0.1, creativity: 0.0, anxiety: 0.0
  },
  "phellandrene": {
    energy: 0.3, focus: 0.2, mood: 0.3, body: 0.0, creativity: 0.2, anxiety: 0.0
  },
  "phytol": {
    energy: -0.4, focus: -0.1, mood: 0.0, body: 0.3, creativity: 0.0, anxiety: -0.3
  },
}
```

4.2 Unknown Terpene Handling

When a terpene appears in inventory COA but is not in `TERPENE_INFLUENCES`:

```
UNKNOWN_TERPENE_COEFFICIENTS = {
  energy: 0.0,
  focus: 0.0,
  mood: 0.0,
  body: 0.0,
  creativity: 0.0,
  anxiety: 0.15  // Conservative anxiety penalty
}

UNKNOWN_TERPENE_CONFIDENCE_PENALTY = 0.1  // Applied to final blend score
```

Rule: Unknown terpenes contribute neutral directional influence but increase anxiety estimate and reduce blend confidence.

4.3 Terpene Normalization

Terpene names in COAs are normalized before lookup:

1. Convert to lowercase
2. Remove whitespace, hyphens, underscores
3. Common alias mapping (e.g., "β-caryophyllene" → "caryophyllene")

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5. CANNABINOID INTENSITY MODELING

5.1 THC Influence (MODELED PARAMETERS)

```
THC_INTENSITY_FACTOR = 0.05  // Per percentage point of THC

THC_MODIFIERS = {
  body: 0.02,      // THC increases body sensation
  anxiety: 0.03,   // THC increases anxiety risk
}
```

5.2 CBD Influence (MODELED PARAMETERS)

```
CBD_ANXIETY_REDUCTION = -0.04  // Per percentage point of CBD
CBD_ENERGY_DAMPING = -0.01     // CBD slightly reduces energy extremes
CBD_THC_BUFFER_RATIO = 0.25    // CBD:THC ratio that significantly buffers anxiety
```

5.3 Cannabinoid Synergy

```
When CBD >= THC * CBD_THC_BUFFER_RATIO:
  anxiety_modifier *= 0.7  // 30% anxiety reduction
```

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6. CONTEXT MODIFIERS (MODELED ADJUSTMENTS)

6.1 Time of Day

```
TIME_MODIFIERS = {
  morning: {
    energy: +0.2,
    focus: +0.1,
    maxAnxiety: -0.05  // Slightly lower anxiety tolerance in morning
  },
  afternoon: {
    energy: 0.0,
    focus: +0.05,
    maxAnxiety: 0.0
  },
  evening: {
    energy: -0.1,
    body: +0.1,
    maxAnxiety: +0.05
  },
  night: {
    energy: -0.3,
    body: +0.2,
    maxAnxiety: +0.1
  }
}
```

6.2 Tolerance

```
TOLERANCE_MODIFIERS = {
  low: {
    thcMultiplier: 0.7,   // Prefer lower THC
    anxietyPenalty: +0.15  // Higher anxiety sensitivity
  },
  medium: {
```

```
    thcMultiplier: 1.0,
    anxietyPenalty: 0.0
  },
  high: {
    thcMultiplier: 1.3,   // Can handle higher THC
    anxietyPenalty: -0.1  // Lower anxiety sensitivity
  }
}
```

6.3 Experience Level

```
EXPERIENCE_MODIFIERS = {
  beginner: {
    maxAnxiety: -0.1,      // Lower anxiety ceiling
    complexityPenalty: 0.05  // Slight preference for simpler profiles
  },
  intermediate: {
    maxAnxiety: 0.0,
    complexityPenalty: 0.0
  },
  expert: {
    maxAnxiety: +0.05,
    complexityPenalty: -0.05  // Slight preference for complex profiles
  }
}
```

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7. CULTIVAR EFFECT CALCULATION

7.1 Base Terpene Effect Vector

For each cultivar, calculate effect vector:

```
For each dimension d in [energy, focus, mood, body, creativity, anxiety]:

  effect[d] = Σ(terpene_percent[t] * TERPENE_INFLUENCES[t][d])
              for all terpenes t in cultivar

  // Normalize by total terpene percentage to get weighted average influence
  total_terpene_percent = Σ(terpene_percent[t]) for all t

  if total_terpene_percent > 0:
    effect[d] = effect[d] / total_terpene_percent
  else:
    effect[d] = 0
```

7.2 Cannabinoid Adjustments

```
// THC contributions
intensity_boost = thcPercent * THC_INTENSITY_FACTOR
effect[body] += thcPercent * THC_MODIFIERS[body]
effect[anxiety] += thcPercent * THC_MODIFIERS[anxiety]

// CBD contributions
effect[anxiety] += cbdPercent * CBD_ANXIETY_REDUCTION
effect[energy] *= (1.0 - cbdPercent * CBD_ENERGY_DAMPING * 0.01)

// CBD:THC synergy
if cbdPercent >= thcPercent * CBD_THC_BUFFER_RATIO:
  effect[anxiety] *= 0.7
```

7.3 Unknown Terpene Penalty

```
unknown_terpene_count = count of terpenes not in TERPENE_INFLUENCES
confidence_penalty = unknown_terpene_count * UNKNOWN_TERPENE_CONFIDENCE_PENALTY
```

---

8. BLEND CALCULATION

8.1 Blend Effect Vector

For a blend of N cultivars with ratios $r_1, r_2, ..., r_n$ where $\Sigma r_i = 1$:

```
For each dimension d:
  blend_effect[d] = Σ(rᵢ * cultivar_effect[i][d]) for i = 1 to N
```

8.2 Blend Constraints

- Minimum 2 cultivars, maximum 3 cultivars
- Each ratio must be >= 0.15 (15% minimum contribution)
- Ratios must sum to 1.0 ± 0.001 (floating point tolerance)

8.3 Ratio Search Space

For 2-cultivar blends:

```
ratios = [(r, 1-r) for r in [0.15, 0.20, 0.25, 0.30, 0.35, 0.40, 0.45, 0.50,
                              0.55, 0.60, 0.65, 0.70, 0.75, 0.80, 0.85]]
```

For 3-cultivar blends:

```
ratios = [(r1, r2, 1-r1-r2)
          for r1 in [0.15, 0.20, 0.25, 0.30, 0.35, 0.40, 0.50, 0.60, 0.70]
          for r2 in [0.15, 0.20, 0.25, 0.30, 0.35, 0.40, 0.50, 0.60, 0.70]
          where r1 + r2 <= 0.85 and (1 - r1 - r2) >= 0.15]
```

---

9. SCORING FUNCTION

9.1 Distance Metric

```
For non-anxiety dimensions (energy, focus, mood, body, creativity):
  dimension_distance[d] = |target[d] - blend_effect[d]|

Euclidean distance:
  effect_distance = sqrt(Σ(dimension_distance[d]²) for d in non-anxiety dimensions)
```

9.2 Anxiety Hard Constraint

```
if blend_effect[anxiety] > intent.constraints.maxAnxiety:
  DISQUALIFY blend (score = -Infinity)
```

9.3 Cannabinoid Constraint Scoring

```
cannabinoid_penalty = 0

blend_thc = Σ(rᵢ * cultivar[i].thcPercent)
blend_cbd = Σ(rᵢ * cultivar[i].cbdPercent)

if minTHC specified and blend_thc < minTHC:
  cannabinoid_penalty += (minTHC - blend_thc) * 0.1

if maxTHC specified and blend_thc > maxTHC:
  cannabinoid_penalty += (blend_thc - maxTHC) * 0.1

if minCBD specified and blend_cbd < minCBD:
  cannabinoid_penalty += (minCBD - blend_cbd) * 0.1

if maxCBD specified and blend_cbd > maxCBD:
  DISQUALIFY blend (hard constraint for safety)
```

9.4 Final Score

```
base_score = -effect_distance  // Negative because we minimize distance

final_score = base_score
              - cannabinoid_penalty
              - confidence_penalty
              - (complexity_penalty if applicable)

where:
  complexity_penalty = 0.05 if 3-cultivar blend and experience=beginner
                       0.0 otherwise
```

---

10. BLEND SELECTION ALGORITHM

10.1 Candidate Generation

```
candidates = []

For each available cultivar C1:
```

```
  For each available cultivar C2 where C2 != C1:
    For each ratio pair (r1, r2) in 2-cultivar search space:
      blend = {cultivars: [C1, C2], ratios: [r1, r2]}
      effect = calculate_blend_effect(blend)
      score = calculate_score(effect, intent)
      if score > -Infinity:  // Passed anxiety constraint
        candidates.append({blend, effect, score})

For each available cultivar C1:
  For each available cultivar C2 where C2 != C1:
    For each available cultivar C3 where C3 != C1 and C3 != C2:
      For each ratio triple (r1, r2, r3) in 3-cultivar search space:
        blend = {cultivars: [C1, C2, C3], ratios: [r1, r2, r3]}
        effect = calculate_blend_effect(blend)
        score = calculate_score(effect, intent)
        if score > -Infinity:
          candidates.append({blend, effect, score})
```

10.2 Ranking and Selection

```
1. Sort candidates by score (descending)
2. In case of tie (scores within 0.0001):
   a. Prefer 2-cultivar over 3-cultivar
   b. If still tied, prefer lower total THC
   c. If still tied, use lexicographic cultivar ID ordering
3. Return top 5 candidates
```

10.3 No-Solution Handling

```
if len(candidates) == 0:
  return {
    error: "NO_VALID_BLEND",
    reason: "No blend satisfies anxiety constraint",
    suggestion: "Increase maxAnxiety or adjust target effects"
  }

if all candidates have cannabinoid violations:
  return {
    error: "CANNABINOID_CONSTRAINTS_UNSATISFIABLE",
    reason: "Available inventory cannot meet THC/CBD requirements"
  }
```

11. OUTPUT SCHEMA

11.1 Blend Recommendation

```typescript
interface BlendRecommendation {
  cultivars: Array<{
    id: string;
    name: string;
    ratio: number;  // [0.15, 1.0], sum to 1.0
  }>;
  predictedEffects: {
    energy: number;
    focus: number;
    mood: number;
    body: number;
    creativity: number;
    anxiety: number;
  };
  cannabinoids: {
    thc: number;
    cbd: number;
  };
  score: number;
  confidence: number;  // [0, 1], reduced by unknown terpenes
  metadata: {
    unknownTerpeneCount: number;
    constraintsViolated: string[];
  };
}

interface EngineOutput {
  recommendations: BlendRecommendation[];  // Top 5, sorted by score
  intent: Intent;  // Echo back for verification
  inventory_timestamp: string;
  calculation_timestamp: string;
}
```

12. DETERMINISM GUARANTEES

12.1 Floating Point

- All calculations use IEEE 754 double precision
- Equality comparisons use epsilon = 0.0001
- Summations performed in consistent order (sorted by cultivar ID)

12.2 Sorting

- Primary: score (descending)
- Tiebreaker 1: cultivar count (ascending)
- Tiebreaker 2: total THC (ascending)
- Tiebreaker 3: lexicographic cultivar IDs (ascending)

12.3 Random Number Generation

FORBIDDEN. No randomness permitted.

12.4 External Dependencies

- No network calls
- No system time (except for audit timestamps in output)
- No file I/O except reading provided inventory

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13. FAILURE MODES

Condition	Response
Intent values out of range	Clamp to valid range, log warning
Inventory empty	Return error: "NO_INVENTORY"
No available cultivars	Return error: "NO_AVAILABLE_CULTIVARS"
No blend satisfies anxiety	Return error: "NO_VALID_BLEND"
Unknown terpene encountered	Apply UNKNOWN_TERPENE_COEFFICIENTS, log terpene name

Condition	Response
Cannabinoid hard constraint violated	Exclude blend from candidates
NaN in calculation	Treat as error, halt, return diagnostic

---

14. VERSIONING AND EXTENSIBILITY

14.1 Configuration Version

All outputs include:

```
{
  "configVersion": "1.0.0",
  "terpeneModelVersion": "1.0.0"
}
```

14.2 Terpene Model Updates

New terpenes added to `TERPENE_INFLUENCES` do not break determinism for existing terpenes.

Previously unknown terpenes move from "unknown" to "known" category.

Recomputation with updated model may yield different results (version mismatch expected).

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15. COMPUTATIONAL COMPLEXITY

- **Cultivars:** N
- **2-cultivar candidates:** N(N-1) × 15 ratios ≈ $15N^2$
- **3-cultivar candidates:** N(N-1)(N-2) × ~81 ratios ≈ $81N^3$
- **Per-candidate cost:** O(T) where T = average terpene count

Typical: N=50, T=10 → ~10M candidates → 100M terpene operations

Optimization note: Pre-compute cultivar effect vectors once before blend enumeration.

---

16. AUDIT AND REPRODUCIBILITY

Every output includes:

```
{
  "audit": {
    "inputHash": "<SHA256 of intent + inventory>",
    "configVersion": "1.0.0",
    "calculationTimestamp": "<ISO 8601>",
    "candidatesEvaluated": <integer>,
    "executionTimeMs": <integer>
  }
}
```

Reproducibility test: Same `inputHash` and `configVersion` → identical `recommendations` array.

END OF SPECIFICATION